Date.....

Week-1 Module

Part 1 :- Star Patterns

① Square

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Logic :- 1) Use two nested loops : an outer loop for rows, an inner loop for columns
2) Outer loop runs from 1 to n
3) Inner loop runs col from 1 to n, printing * each time
4) After inner loop finishes, print a newline to move to the next row.

Code :-
```
public class Square {
    public static void main (String [] args) {
        int n = 5;
        for (int row = 1; row ≤ n; row++) {
            for (int col = 1; col ≤ n; col++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

② Hollow Square

```
* * * * *
*       *
*       *
*       *
* * * * *
```

Logic :-
1) Same nested loop, an outer loop for rows and inner loop for columns.
2) for each cell (row, col) print * only if its on the border - i.e, row = 1, row = n, col = 1 or col = n.
3) Otherwise, print a space, so inside stays hollow.

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Code :- public class HollowSquare {
public static void main (String [] args) {
int n = 5;
for (int row = 1; row ≤ n ; row ++) {
for (int col = 1 ; col ≤ n ; col ++) {
if (row = 1 || row = n || col = 1 || col = n) {
System. out. print (" * ");
} else {
System. out. print (" ");
}
}
System. out. println ( );
}
}

(3) Right Triangle

```
*
* *
* * *
* * * *
* * * * *
```

Logic :-
1) Outer loop runs from 1 to n.
2) Inner loop runs col from 1 to row (not to n!) - so row 1 prints 1 star, row 2 prints 2 stars and so on.
3) Print a newline after each row.

Code :- public class Triangle {
public static void main (String [] args) {
int n = 5;
for (int row = 1; row ≤ n; row ++) {
for (int col = 1; col ≤ n; col ++) {
System. out. print ("*");
}
System. out. println ( );
}
}

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④ Inverted Triangle

```
* * * * *
* * * *
* * *
* *
*
```

Logic :-

1) Outer loop runs row from 1 to n.
2) Inner loop runs col from 1 to (n-row+1) - this starts at n stars and shrinks by 1 each row.
3) Print a newline after each row

Code :-

```
public class InvertedTriangle {
    public static void main (String [] args) {
        int n=5;
        for (int row = 1; row ≤ n ; row++) {
            for (int col = 1; col ≤ n-row+1; col++) {
                System.out.print ("*");
            }
            System.out.println (" ");
        }
    }
}
```

⑤ Pyramid

```
    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```

Logic :-

1) Outer loop runs row from 1 to n.
2) First, print (n-row) leading spaces to push stars towards center.
3) then print (2*row-1) stars - this gives odd sequence 1,3,5,7,9 and print newline.

Code :-

```
public class Pyramid {
    public static void main (String [] args) {
        int n=5;
        for (int row=1; row ≤ n ; row++) {
            for (int sp = 1; sp ≤ n-row; sp++) {
                System.out.print (" ");
            }
```

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```
for (int star=1; star <= 2*row-1; star++) {
        System.out.print("*");
    }
    System.out.println();
}
}
```

⑥ Diamond

```
    *
   * * *
  * * * * *
 * * * * * * *
  * * * * *
   * * *
    *
```

Logic :-

1) A diamond is a pyramid (growing half) immediately followed by upside down pyramid ↳ 9 rows total for n=5.
2) Use one loop variable i from 1 to (2n-1)
3) Compute a level K if i ≤ n, then K=i otherwise K = 2n-i (shrinking half)
4) Print (n-K) spaces, then (2K-1) stars - exactly the same formula as pyramid, driven by K.

Code :-

```
public class Diamond {
    public static void main (String [] args) {
        int n=5;
        for (int i=1; i<= n*2-1; i++) {
            int K= (i<=n) ? i: 2*n-i;
            for (int sp=1; sp <= n-k; sp++) {
                System.out.print(" ");
            }
            for (int star=1; star <= 2*K-1; star++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

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⑦ Half - Diamond

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

Logic :-

1) Same "grow then shrink" idea as diamond but left-aligned and increasing by 1 star per row instead of 2.
2) Use i from 1 to (2n-1), with the same level formula : K = (i ≤ n) ? i : 2n - i
3) Print K stars (no leading spaces needed since it's flush out).

Code :-

```
public class HalfDiamond {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= 2 * n - 1; i++) {
            int k = (i <= n) ? i : 2 * n - i;
            for (int star = 1; star <= k; star++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

⑧ Half Diamond Inverted

```
* * * * *
* * * *
* * *
* *
*
```

Logic :-

1) Same logic as Inverted Triangle : outer loop row from 1 to n
2) Inner loop prints (n - row + 1) stars, decreasing each row.

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Code :-
```
public class HalfDiamondInverted {
    public static void main (String [] args) {
        int n=5;
        for (int row = 1; row ≤ n ; row++){
            for (int star = 1; star ≤ n - row + 1; star++){
                System.out.print ("*");
            }
            System.out.println ();
        }
    }
}
```

Part 2:- Number Patterns

① Square

```
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

Logic :-

1) Same nested-loop as the star square
2) Outer loop row from 1 to n, inner loop col from 1 to n.
3) Instead of printing *, print digit "1" every time.

Code :-
```
public class NumberSquare {
    public static void main (String [] args) {
        int n=5;
        for (int row = 1; row ≤ n ; row++){
            for (int col = 1; col ≤ n ; col++){
                System.out.print ("1");
            }
            System.out.println ();
        }
    }
}
```

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② Incrementing

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

Logic :-
1) Outer loop row from 1 to n - every row looks identical.
2) Inner loop col from 1 to n, printing the value of col each time.

Code :-
```
public class Incrementing {
    public static void main (String [] args) {
        int n = 5;
        for (int row = 1; row ≤ n; row++) {
            for (int col = 1; col ≤ n; col++) {
                System.out.print (col + " ");
            }
            System.out.println ();
        }
    }
}
```

③ Right Triangle

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Logic :-
1) Outer loop row from 1 to n
2) Inner loop col from 1 to row, printing the value of col each time - so each row counts up to its row number.

Code :-
```
public class RightTriangle {
    public static void main (String [] args) {
        int n = 5;
        for (int row = 1; row ≤ n; row++) {
            for (int col = 1; col ≤ row; col++) {
                System.out.print (col + " ");
            }
            System.out.println ();
        }
    }
}
```

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(4) Inverted Diamond

```
1 1 1 1 1
2 2 2 2
3 3 3
4 4
5
```

Logic :-

1) Outer loop row from 1 to n.
2) The digit printed on row is (row+2)
3) The no. of times it printed is (n - row + 1)
4) Inner loop runs col from 1 to count, printing value each time.

Code :-

```
public class InvertedDiamond {
    public static void main (String [] args) {
        int n = 5;
        for (int row = 1; row ≤ n; row++) {
            int value = row + 2;
            int count = n - row + 1;
            for (int col = 1; col ≤ count; col++) {
                System.out.print (value + " ");
            }
            System.out.println();
        }
    }
}
```

(5) Sandwich Pattern

```
3 3 3 3 3
  4 4 4
    5
  4 4 4
3 3 3 3 3
```

Logic :-

1) Outer loop row from 1 to n.
2) Find how far the current row is from the nearer edge : distance = min (row-1, n-row)
3) count = n - 2 * distance (shrinks towards middle)
4) value = (n+1)/2 + distance - the digit to print.
5) Print distance * 2 leading spaces, then print value repeated count times

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Code :-

```
public class SandwichPattern {
    public static void main(String [] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            int distance = Math.min(row - 1, n - row);
            int count = n - 2 * distance;
            int value = (n+1)/2 + distance;
            for (int sp = 1; sp <= distance * 2; sp++) {
                System.out.print(" ");
            }
            for (int col = 1; col <= count; col++) {
                System.out.print(value + " ");
            }
            System.out.println();
        }
    }
}
```

⑥ Incrementing Diamond

```
    3
   4 4
  5 5 5
 6 6 6 6
7 7 7 7 7
 6 6 6 6
  5 5 5
   4 4
    3
```

Logic :-

1) Same "grow then shrink" shape as the star diamond - 9 rows for n = 5
2) Use i from 1 to (2n - 1), with level K = (i <= n) ? i : 2n - i (gives 1,2,3,4,5,4,3,2,1)
3) The digit printed is value = K + 2, where K = n, print largest digit.
4) Print (n - K) * 2 leading spaces to center the row, print value repeated K times.

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Code :-

```
public class IncrementingDiamond {
    public static void main (String [] args) {
        int n = 5;
        for (int i = 1; i <= 2 * n - 1; i++) {
            int k = (i <= n) ? i : 2 * n - i;
            int value = k + 2;
            for (int sp = 1; sp <= (n - k) * 2; sp++) {
                System.out.print (" ");
            }
            for (int col = 1; col <= k; col++) {
                System.out.print (value + " ");
            }
            System.out.println();
        }
    }
}
```

Square.java

Square.java > Language Support for Java(TM) by Red Hat > Square > main(String[])

```java
public class Square {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int col = 1; col <= n; col++) {
                System.out.print(s: "*");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powersh... Run: Square

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'Square'
*****
*****
*****
*****
*****
PS D:\oops with java>
```

HollowSquare.java

HollowSquare.java > Java > HollowSquare

```java
public class HollowSquare {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int col = 1; col <= n; col++) {
                if (row == 1 || row == n || col == 1 || col == n) {
                    System.out.print(s: "*");
                } else {
                    System.out.print(s: " ");
                }
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: HollowSquare

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'HollowSquare'
*****
*   *
*   *
*   *
*****
PS D:\oops with java>
```

Triangle.java

```java
public class Triangle {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int col = 1; col <= row; col++) {
                System.out.print(s: "*");
            }
            System.out.println();
        }
    }
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: Triangle

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'Triangle'
*
**
***
****
*****
PS D:\oops with java>
```

Triangle.java | InvertedTriangle.java

InvertedTriangle.java > Java > InvertedTriangle

```java
public class InvertedTriangle {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int col = 1; col <= n - row + 1; col++) {
                System.out.print(s: "*");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: InvertedTriangle

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'InvertedTriangle'
*****
****
***
**
*
PS D:\oops with java>
```

Pyramid.java

Pyramid.java > Java > Pyramid

```java
public class Pyramid {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int sp = 1; sp <= n - row; sp++) {
                System.out.print(s: " ");
            }
            for (int star = 1; star <= 2 * row - 1; star++) {
                System.out.print(s: "*");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: Pyramid

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'Pyramid'
    *
   ***
  *****
 *******
*********
PS D:\oops with java>
```

Diamond.java

Diamond.java > Java > Diamond > main(String[] args)

```java
public class Diamond {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= 2 * n - 1; i++) {
            int k = (i <= n) ? i : 2 * n - i;
            for (int sp = 1; sp <= n - k; sp++) {
                System.out.print(s: " ");
            }
            for (int star = 1; star <= 2 * k - 1; star++) {
                System.out.print(s: "*");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: Diamond

```
PS D:\oops with java>  d:; cd 'd:\oops with java'; & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'Diamond'
    ***
   *****
  *******
 *********
  *******
   *****
    ***
     *
PS D:\oops with java>
```

HalfDiamond.java

```java
public class HalfDiamond {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= 2 * n - 1; i++) {
            int k = (i <= n) ? i : 2 * n - i;
            for (int star = 1; star <= k; star++) {
                System.out.print(s: "*");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: HalfDiamond

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'HalfDiamond'
****
*****
****
***
**
*
PS D:\oops with java>
```

Incrementing.java

```java
public class Incrementing {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int col = 1; col <= n; col++) {
                System.out.print(col + " ");
            }
            System.out.println();
        }
    }
}
```

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'Incrementing'
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
PS D:\oops with java>
```

RightTriangle.java

```java
public class RightTriangle {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            for (int col = 1; col <= row; col++) {
                System.out.print(col + " ");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: RightTriangle

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'RightTriangle'
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
PS D:\oops with java>
```

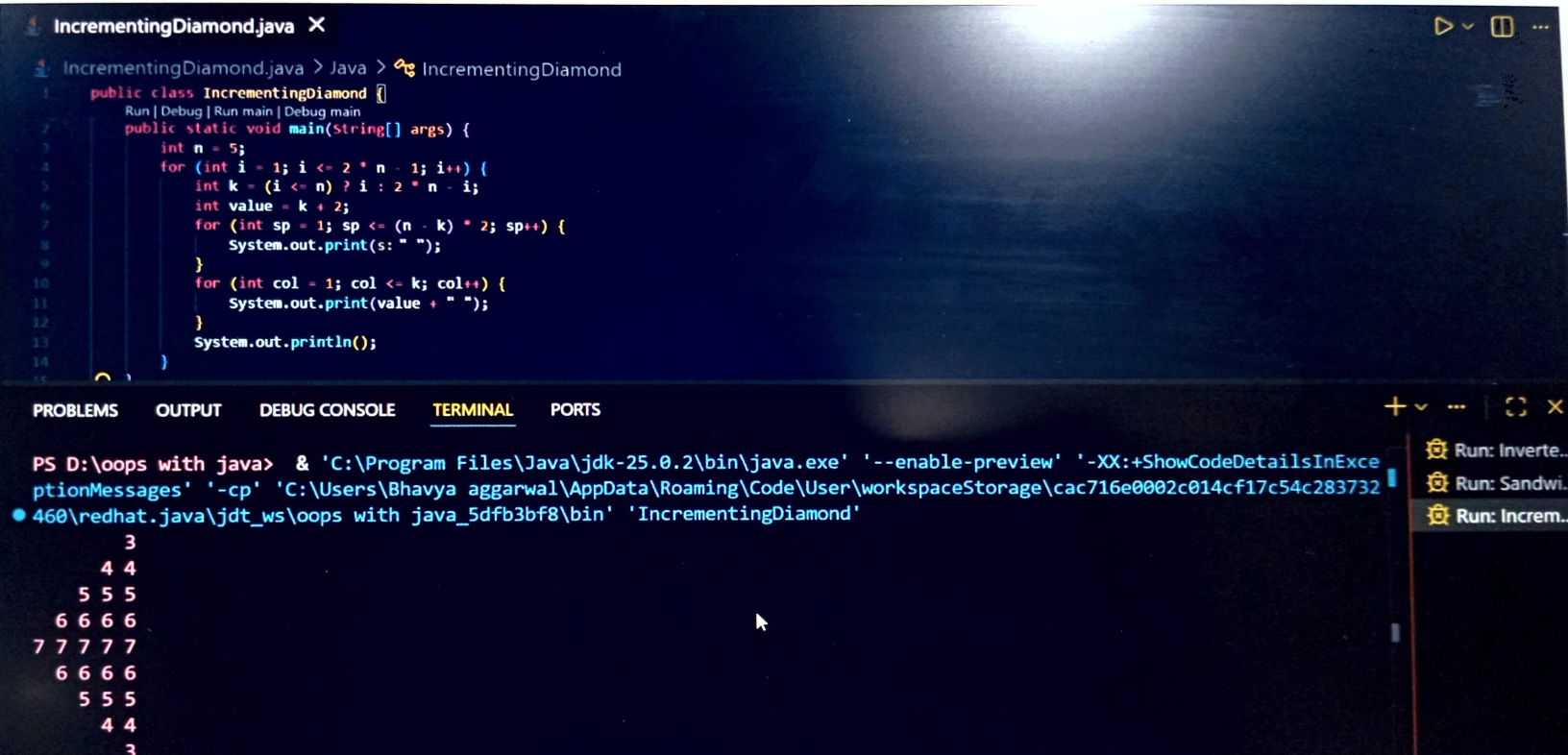
SandwichPattern.java

```java
public class SandwichPattern {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        int n = 5;
        for (int row = 1; row <= n; row++) {
            int distance = Math.min(row - 1, n - row);
            int count = n - 2 * distance;
            int value = (n + 1) / 2 + distance;
            for (int sp = 1; sp <= distance * 2; sp++) {
                System.out.print(s: " ");
            }
            for (int col = 1; col <= count; col++) {
                System.out.print(value + " ");
            }
            System.out.println();
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\oops with java>  & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'SandwichPattern'
3 3 3 3 3
  4 4 4
    5
  4 4 4
3 3 3 3 3
PS D:\oops with java>
```

Run: Inverte...
Run: Sandwi...


IncrementingDiamond.java
IncrementingDiamond.java > Java > IncrementingDiamond
public class IncrementingDiamond {
Run | Debug | Run main | Debug main
public static void main(String[] args) {
int n = 5;
for (int i = 1; i <= 2 * n - 1; i++) {
int k = (i <= n) ? i : 2 * n - i;
int value = k + 2;
for (int sp = 1; sp <= (n - k) * 2; sp++) {
System.out.print(s: " ");
}
for (int col = 1; col <= k; col++) {
System.out.print(value + " ");
}
System.out.println();
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS D:\oops with java> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Bhavya aggarwal\AppData\Roaming\Code\User\workspaceStorage\cac716e0002c014cf17c54c283732460\redhat.java\jdt_ws\oops with java_5dfb3bf8\bin' 'IncrementingDiamond'
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
Run: Inverte...
Run: Sandwi...
Run: Increm...